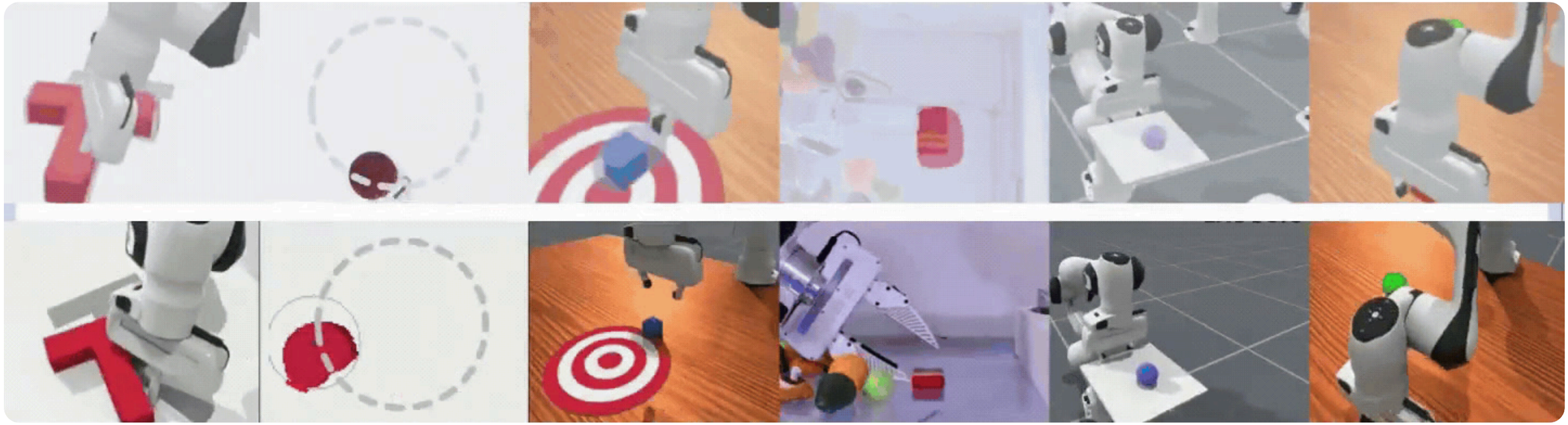
ManiDreams

RICE UNIVERSITY

An Open-Source Library for Robust Object Manipulation via Uncertainty-aware Task-specific Intuitive Physics



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Your world model predicts
a successful push.

Would you press execute?

Now change the friction.

The robot must act before it knows.



Illustrative simulation from the project video

Uncertainty reaches the planner

A real action has to work with imperfect information.



Perceptual

Where is the object right now?

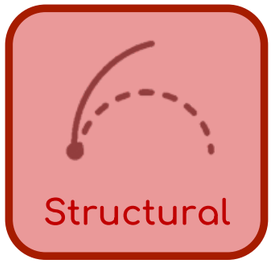
Noisy or delayed observations



Parametric

What physics will this contact have?

Mass, friction and geometry



Structural

How well does the model describe it?

Imperfect dynamics models

Carry plausible outcomes into action selection.

One object, several plausible states

Dream it · Domain-Randomized Instance Set (DRIS)



Multiple hypotheses about **one target**.
Different states and physical contexts.

$$\mathcal{D}_t = \left\{ \left(s_t^{(i)}, c^{(i)} \right) \right\}_{i=1}^{m}$$

state s / physical context c / m instances

Different actions, different possible outcomes

Predict it · Task-specific intuitive physics (TSIP)

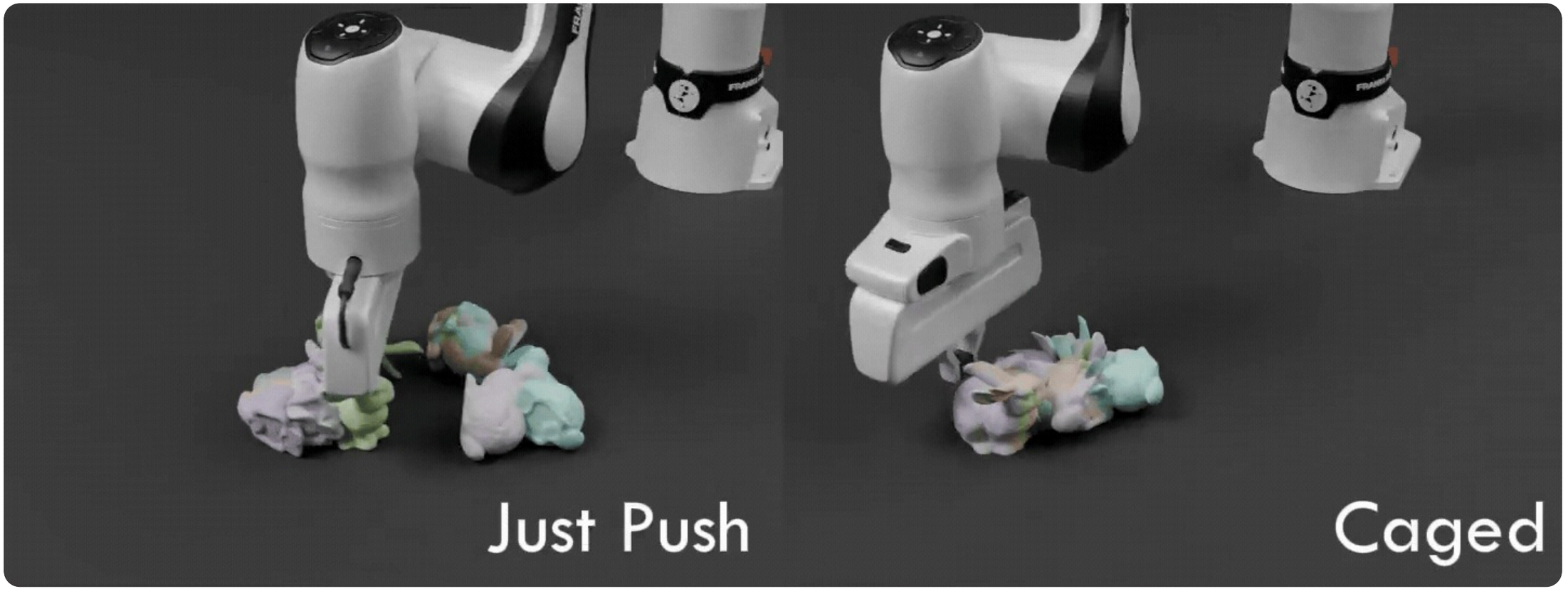


TSIP predicts the next DRIS
from the current set and an action.

Which predicted spread
should the task accept?

Which outcomes does the task accept?

Constrain it · Cage evaluation and validation

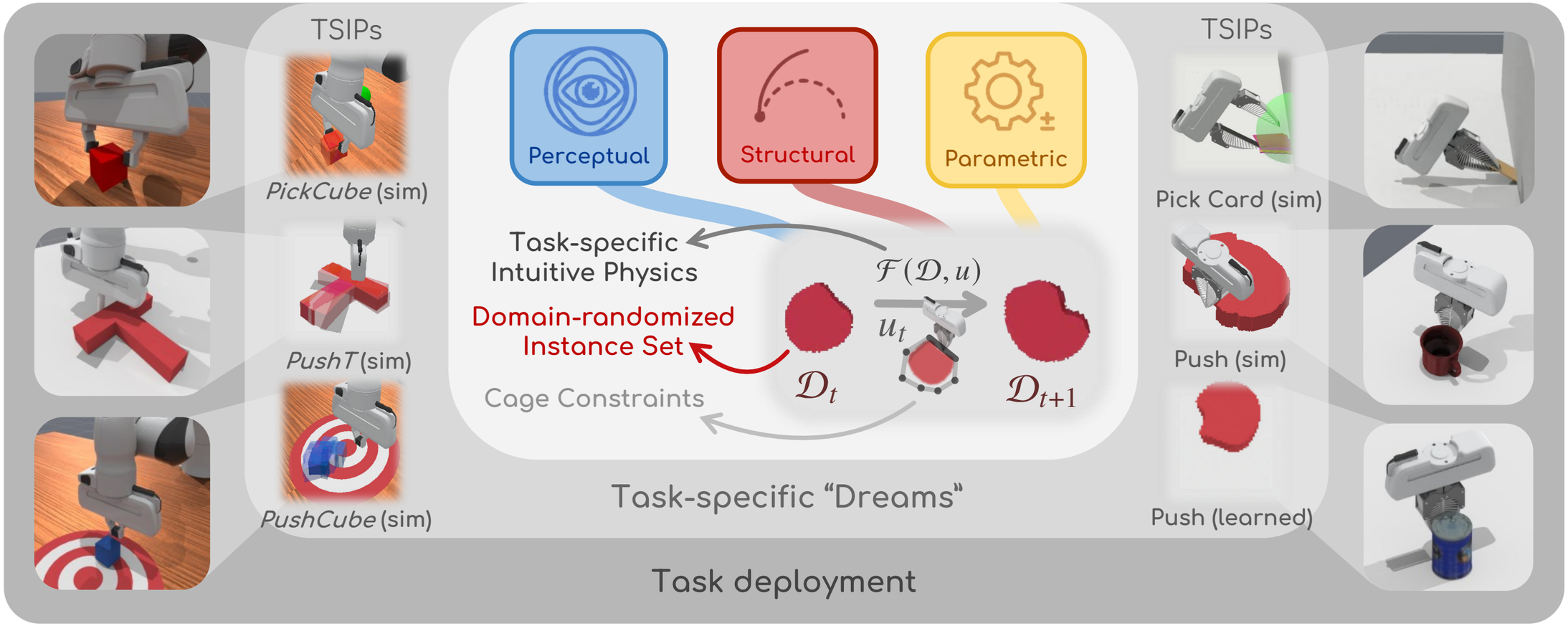


evaluate() scores the outcome.
validate() checks containment.

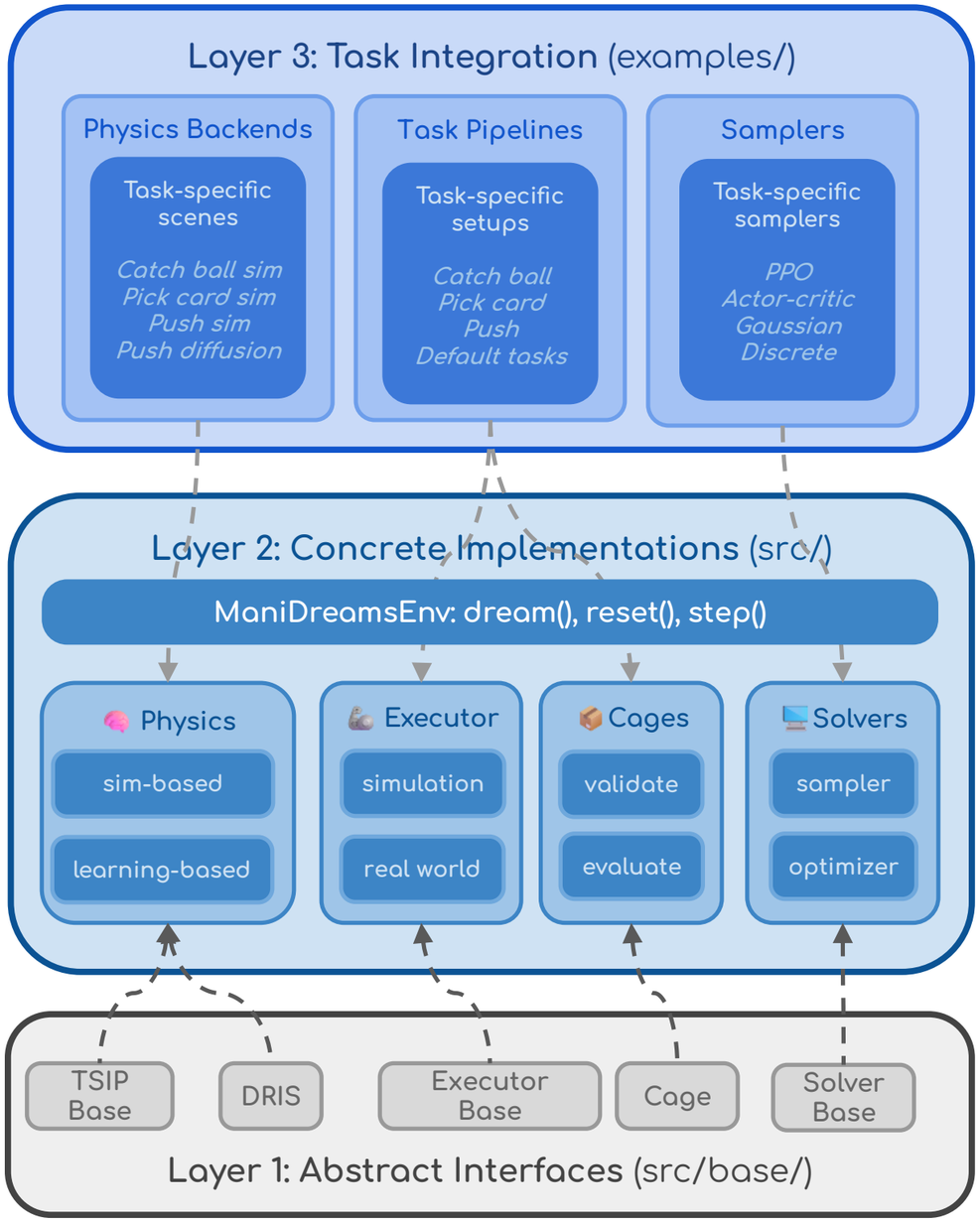
Qualitative simulation illustration. The Solver uses both to select an action.

Uncertainty inside the planning loop

A common representation connects different tasks and physics backends.



Robust manipulation is an integration problem.



A layered framework for manipulation

Separate task definitions, component implementations and interfaces.

Task integration

Supply a scene, pipeline configuration and action sampler.

Concrete implementations

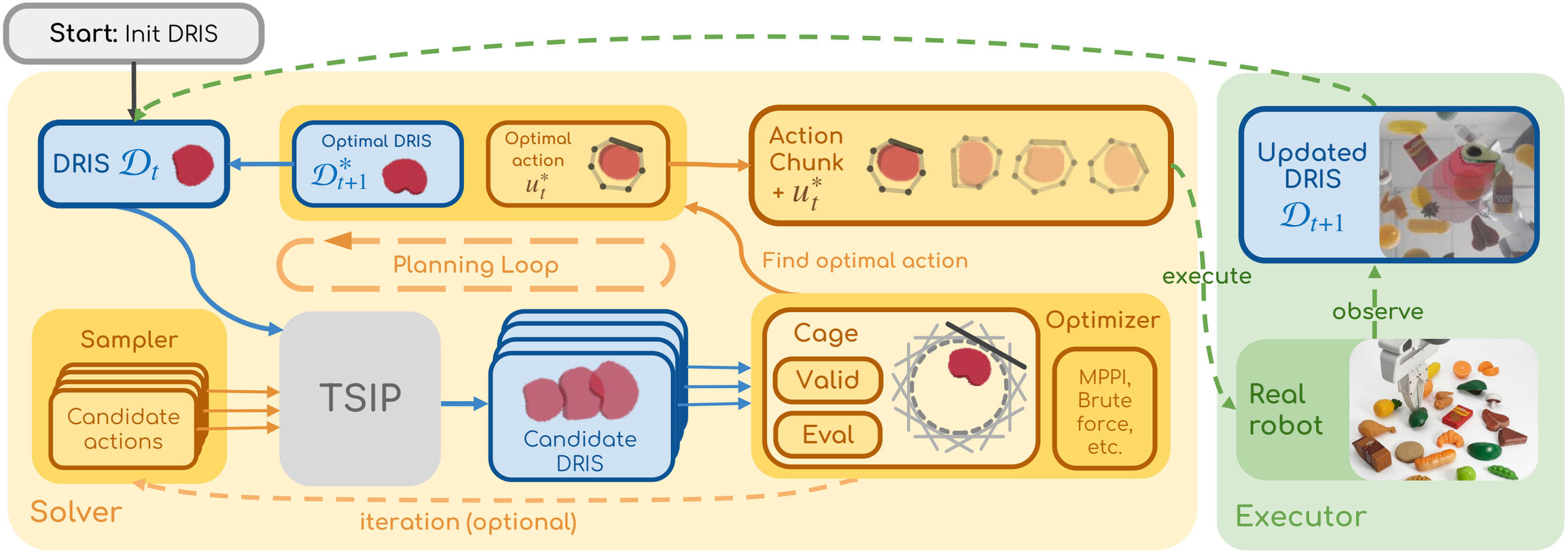
Components share ManiDreamsEnv.

Abstract interfaces

DRIS, TSIP, Cage, Solver and Executor

Together, they form a planning loop

Four abstractions wrap the base policy without retraining.



1 Sample

Solver proposes N actions

2 Predict

TSIP propagates each DRIS

3 Constrain

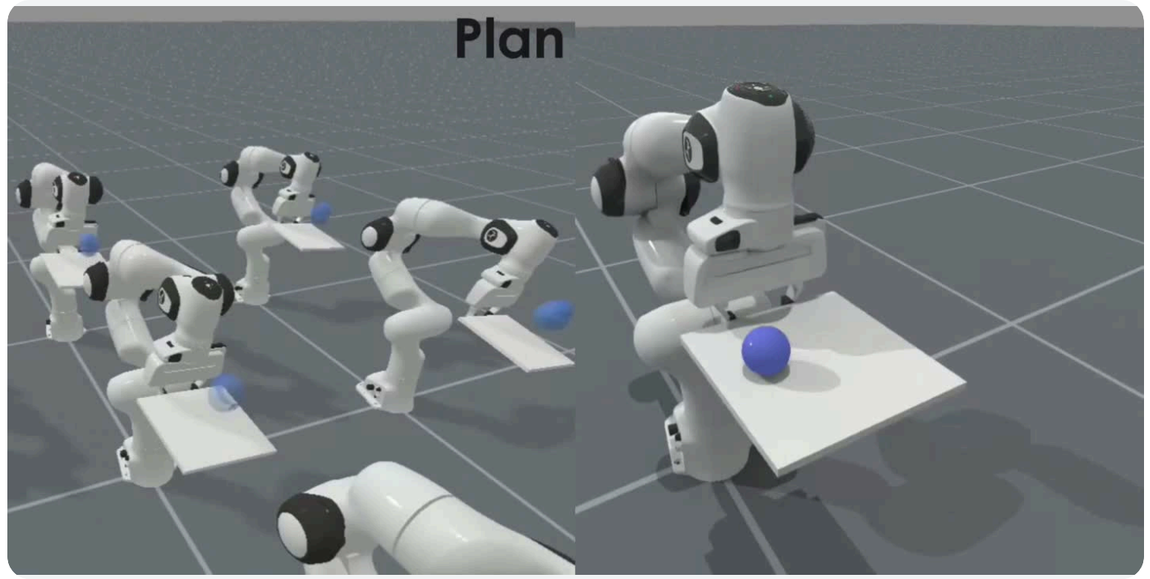
Cage scores and validates

4 Execute

Act, observe, refresh DRIS

The same interfaces support different tasks

Changing the sampler, cage and optimizer.



Ball catching

Pretrained PPO sampler

Simulation TSIP



Card picking

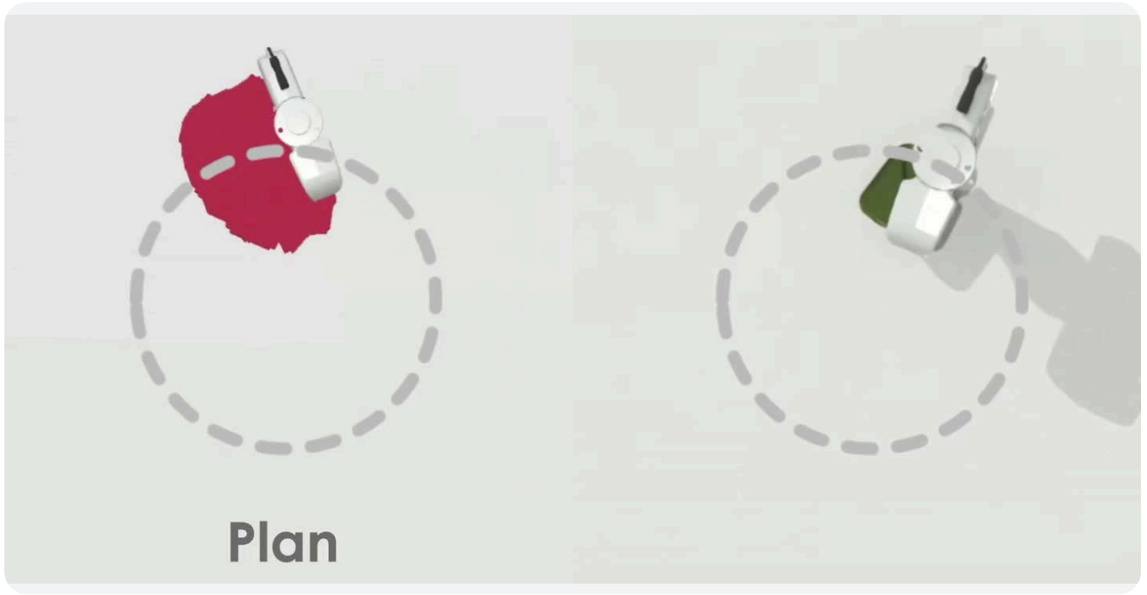
Geometric 3D cage

Continuous sampling + MPPI

One planning interface, independently replaceable components.

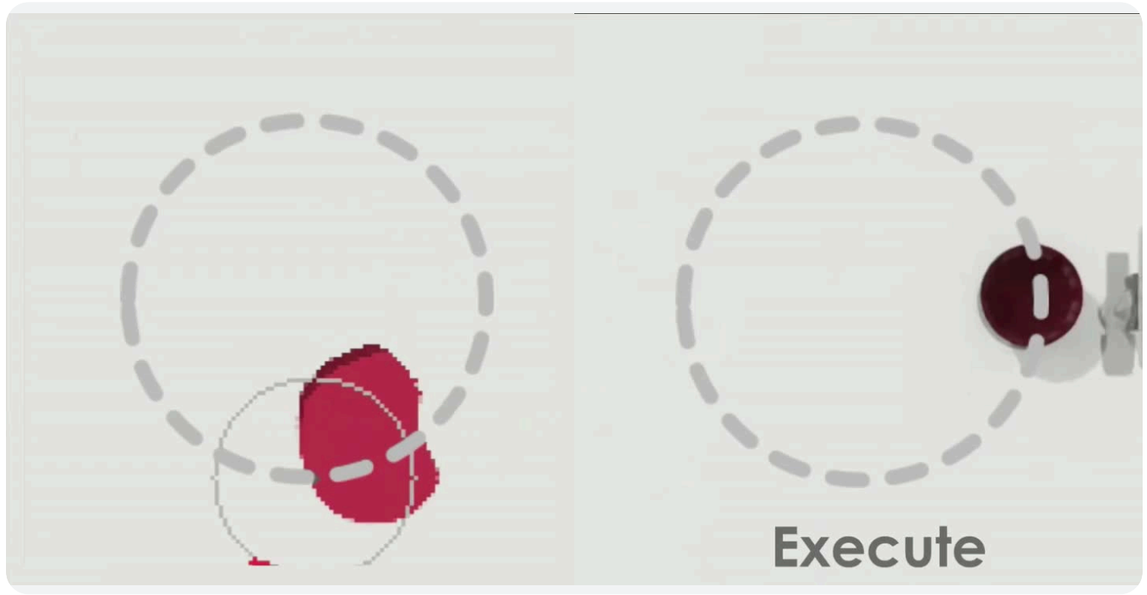
Two physics backends, the same task

Circular object pushing; each pair shows plan (left) and execution (right).



Simulation-based TSIP

Parallel physical instances



Diffusion-based TSIP

Prediction over a DRIS representation

Unchanged Cage Discrete sampler N-best solver

The loop reaches the physical robot

Franka Panda · diffusion TSIP · SAM2 pixel DRIS

Push for clearance, then pick

Croissant



Chicken leg



Use a wall corner to scoop



Steak



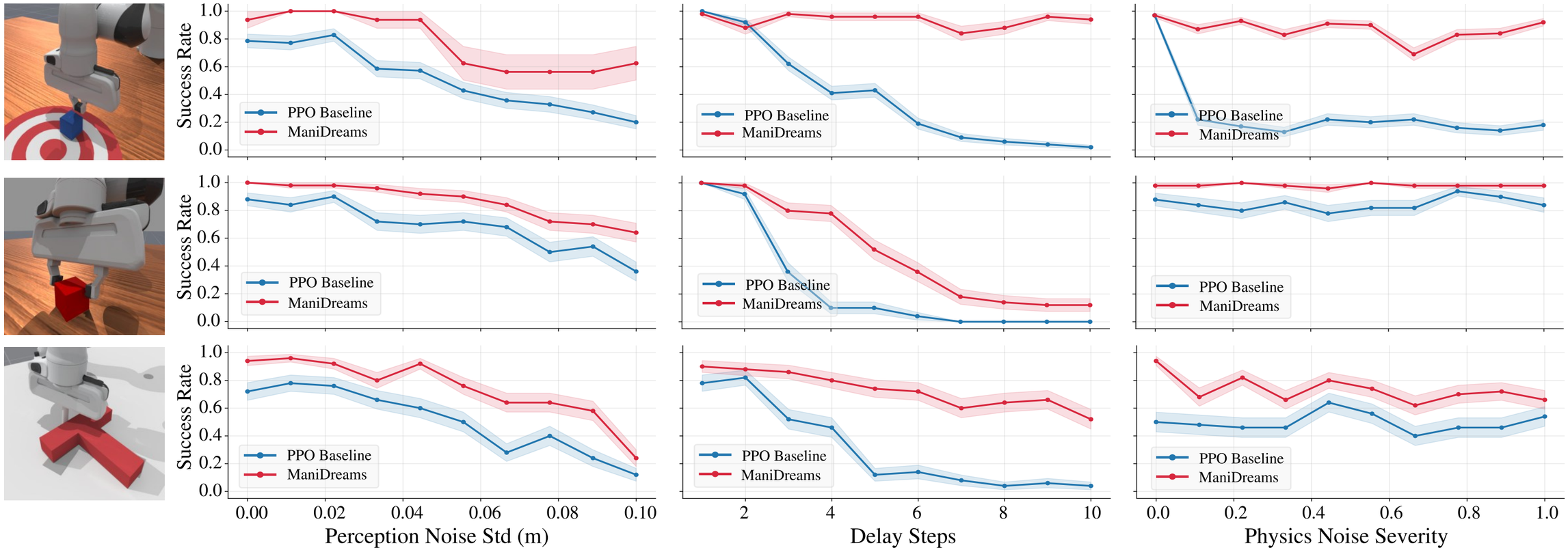
Thin box

8-action chunks. Observe and refresh between chunks.
Zero-shot executor swap, without fine-tuning.

Camera view from beneath a transparent table.
Qualitative demonstrations; original x2 playback.

The same PPO policy under perturbation

PPO: direct execution ManiDreams: PPO as the sampler



Rows: PushCube, PickCube, PushT. $N = m = 8$. Each perturbation swept independently. 10 runs $\times$100 episodes; $\pm$1 SD.

Largest gains at moderate perturbations.

At extremes, the fixed DRIS can stop covering the uncertainty.

Coverage and compute set the trade-off

PushCube ablations and measured runtime

Success under combined perturbations

m or N	1	4	8	16	32
Vary DRIS size m	58	72	**82**	85	86
Vary samples N	52	71	**82**	87	88

74%
Narrow

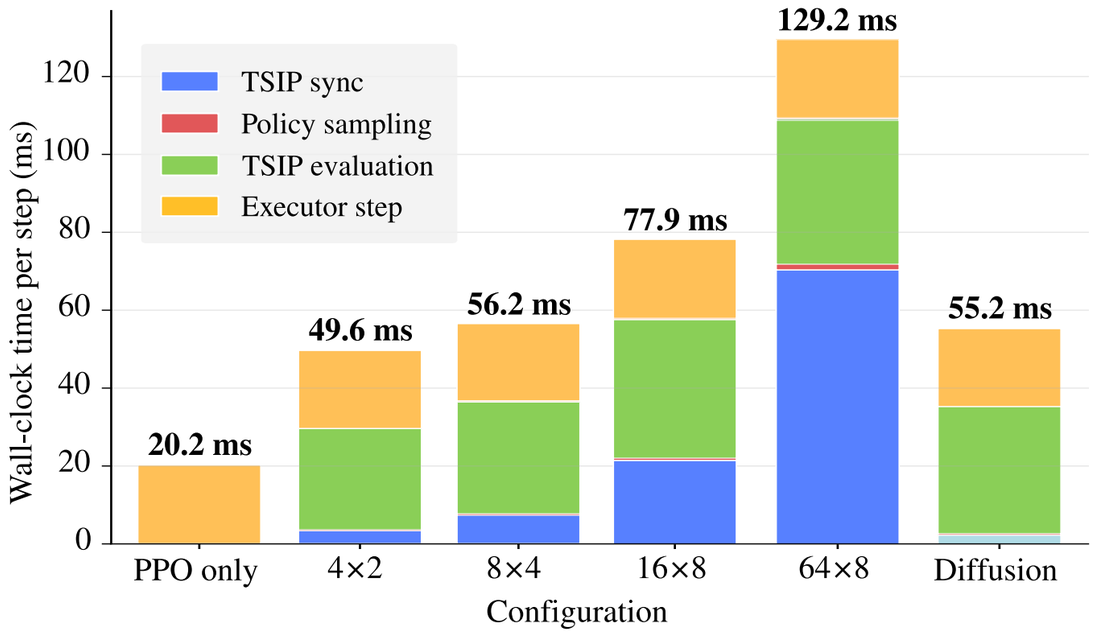
82%
Medium

76%
Wide

Diminishing returns above 16.
Medium width works best in this setting.

Noise $\sigma = 0.04\,\mathrm{m}$; chunk length 3; physics severity 0.4.
100 episodes/config. Others held at $m = N = 8$, Medium.

Wall-clock time per step



20.2 ms PPO / 55.2 ms diffusion TSIP

Single NVIDIA RTX 5070 Ti laptop GPU.
Simulation configurations: 49.6–129.2 ms/step.

Which futures should the robot accept?

Keep uncertainty inside the planning loop.
Add robustness around the policy, without retraining.

Current limits: task-specific DRIS and cage tuning,
and a suitable dynamics backend.

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Explore ManiDreams

Project · Demos · Slides · Code

Evaluation settings

Backup

Robustness benchmark

Same PPO policy for both methods.
$N = 8,\ m = 8,\ \lambda = 0.1.$

Vary one perturbation at a time.
Hold the other two at zero.

10 runs $\times 100$ episodes/configuration.
Report mean and ± 1 standard deviation.

Ablation

PushCube, 100 episodes/configuration.
Noise $\sigma = 0.04\,\mathrm{m}$, chunk length 3,
physics severity 0.4.

Width	Parameter multiplier
Narrow	$[0.9,\ 1.2]\times$ default
Medium	$[0.5,\ 2]\times$ default
Wide	$[0.2,\ 3]\times$ default

Scope and limitations

Backup

Tested perturbations

Observation noise, observation delay and physical parameters.

Manual specification

DRIS parameters and cages need task-specific tuning.

Model dependence

Discontinuous dynamics may require more expressive TSIP backends.

Real-robot evidence

Qualitative deployment demonstrations, without reported success rates.